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An Analysis of Michigan Charter Schools: Enrollment, Revenues, and Expenditures*

Oded Izraeli and Kevin Murphy

ABSTRACT

Though charter schools are a relatively new phenomenon in American education, they have been in existence long enough now to allow for evaluation of operational outcomes relative to traditional public schools. Using a comprehensive data set for the State of Michigan that spans the years 1995 through 2006, this article aims to provide such an evaluation in several dimensions. Specifically, we address four questions with our dataset: (1) To what degree do charter schools compete with local public schools? (2) Are charter schools underfunded relative to public schools? (3) What is the effect on state education finances of operating a system of charter schools? (4) Judging from expenditure from revenue patterns, do charter schools behave more as profit maximizing entities than do public schools?

INTRODUCTION

The education reform movement in the United States evolved as response to a perceived crisis in American public education. According to critics and reformers, neighborhood public schools act as local monopolists and—as monopolists often do—have produced a substandard product in recent years as evidenced by high dropout rates and inadequately prepared college freshmen. According to reformers, the solution is a strong dose of competition in order to induce teachers and students to work harder and to spur high achievement levels. This reform movement can be likened to the proverbial “three-legged stool,” resting on legs consisting of accountability (testing), school choice, and

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charter schools. The focus of this article is on the third leg of the reform stool—charter schools.¹

Charter schools constitute a relatively new phenomenon in American education. Proponents argue that charter schools provide a number of benefits. For example, charter schools provide students—particularly students in failing school districts—legitimate alternatives to their local public school. Moreover, as school districts become more diverse over time, educational preferences may also become more diverse, and charter schools may have arisen to accommodate such growing demand for diversity in educational choices within a public schools framework. As Stoddard and Corcoran (2007, p. 32) explain: “Many charter school founders explicitly state the goal of satisfying diverse preferences and attracting a target student population as central to their mission.” Glomm, Harris, and Lo (2005, p. 454) find strong support for the hypothesis that “... charter schools should be more likely to locate in districts whose populations are more diverse.” In addition, charter schools provide teachers a less bureaucratic and more flexible work environment, promoting teacher initiative, creativity, and innovation. Charter schools, though technically public, have many characteristics of private-sector business and, therefore, may be more likely to adopt efficient management practices. Finally, it is argued that charter schools place competitive pressure on local public schools, thus, generating a positive externality by leading to improved academic performance in traditional public schools.

Review of the relevant literature suggests that, to date, not much work has been done on other aspects of operations related to charter schools. Miron and Nelson (2002, p. 68) find that charter schools operate less efficiently than public schools. Dee and Fu (2004, p. 260) raise the concern that “...the introduction of charter schools...lowered the amount of resources available to conventional public schools.” On the other hand, studies from Gronberg and Janson (2001), Hoxby (2004), Hoxby and Rockoff (2004), Bifulco and Ladd (2006), and Walberg (2007) raise the concern that charter schools are not getting equal funding per student from the state when compared with traditional public schools. No attempt is made, however, in the aforementioned studies to collect information on additional funds available to charter schools (and public schools) from other

1. While the notion of injecting competition in education is a seductive analogy, even capturing the fancy of the Obama Administration, the track record that reform efforts such as No Child Left Behind (NCLB) and various state level initiatives have established over the last decade leaves much to be desired. For example, Nichols, Glass, and Berliner (2005) and Glass (2008) characterize high stakes testing as an abject failure with many adverse unintended consequences. Diane Ravitch, a leading education historian and, as Assistant Secretary of Education under the first President Bush, an ardent supporter of choice and accountability, now repudiates many aspects of education reform as doing much more harm than good (Ravitch 2010).

sources (local and federal government), nor do these studies provide any analysis of developments and trends over time.

The focus in this article centers on some of the operational issues identified above. In particular, we explore four questions: (1) To what degree do charter schools compete with local public schools? (2) Are charter schools underfunded, *ceteris paribus*, in comparison to traditional public schools? (3) What is the effect on state education finances of operating an essentially second system of public education? (4) Do charter schools, judging by expenditure from revenue patterns, act more like profit maximizing businesses than public-good providers?

In conducting our empirical analyses we utilize data on charter schools in the state of Michigan.² The first charters in Michigan commenced operation in 1994. Though modest in scope at the outset—there were three charter schools in 1994—the number of enrollments in Michigan charters increased rapidly in the ensuing years. The average annual growth rate of charter school enrollments during our study period (1996–2006) was almost 29% and the average annual growth rate in the number of charter schools during this period was 15.6%.³ To place these figures in context, there was virtually no change in Michigan's population of school age children (ages 5–18) between 1994 and 2007.⁴

BACKGROUND

Trends in Enrollments

Charter schools were introduced in Michigan in academic year 1994–1995 under the auspices of Public Act 362. The Act was shepherded through the legislature by then Governor, John Engler—an ostensible education reformer. As previously mentioned, Michigan charter schools were few at the outset, but the number of and enrollments in charter schools grew rapidly over the course of our study window—particularly in the latter 1990s (Table 1). By period end, charter enrollments stood at more than 90,000 students.

A central and also basic question is, what degree do traditional public schools *compete* with charter schools for students? In other words, is it the case that the superiority of the charter school model has caused a large-scale exodus of

2. Michigan, under the leadership of former Governor, John Engler, is a recognized pioneer in the charter school movement. According to the Center for Education Reform (2006), the State of Michigan received a grade of "A" for its charter schools law and is ranked fifth among the 41 states that have passed charter schools law.

3. These rates are tabulated from data described in this article. Years 1994 and 1995 are excluded from the calculations in order to avoid unduly overstating the growth rates.

4. Total enrollments in Michigan increased by almost 53,000 students on a base of 1.76 million students during the 1996–2006 period, which translates into an average annual rate of increase of 1/10 of 1%.

Table 1. Enrollments and Number of Schools by School Type.

Year	Charter Schools				Non-Public Schools			
	Enrollments		Schools		Enrollments		Schools	
	Number	Δ	Number	Δ	Number	Δ	Number	Δ
1995	160		3		176,951		1,080	
1996	4,449	4,289	44	41	179,933	2,982	1,072	-8
1997	11,370	6,921	78	34	177,395	-2,538	1,068	-4
1998	21,175	9,805	108	30	177,740	345	1,059	-9
1999	33,298	12,123	138	30	175,412	-2,328	1,059	0
2000	46,833	13,535	171	33	173,602	-1,810	1,060	1
2001	55,894	9,061	184	13	169,701	-3,901	1,049	-11
2002	64,103	8,209	190	6	165,758	-3,943	1,021	-28
2003	64,355	252	189	-1	157,319	-8,439	997	-24
2004	73,768	9,413	191	2	148,667	-8,652	961	-36
2005	81,703	7,935	208	17	142,891	-5,776	931	-30
2006	90,596	8,893	217	9	134,798	-8,093	896	-35
Total Change, 1995 to 2006	90,436		214		-42,153		-184	

Year	Public Schools				All Schools			
	Enrollments		Schools		Enrollments		Schools	
	Number	Δ	Number	Δ	Number	Δ	Number	Δ
1995	1,581,079		3,280		1,758,190		4,363	
1996	1,602,785	21,706	3,343	63	1,787,167	28,977	4,459	96
1997	1,620,931	18,146	3,406	63	1,809,696	22,529	4,552	93
1998	1,627,269	6,338	3,364	-42	1,826,184	16,488	4,531	-21
1999	1,626,268	-1,001	3,391	27	1,834,978	8,794	4,588	57
2000	1,617,010	-9,258	3,461	70	1,837,445	2,467	4,692	104
2001	1,612,393	-4,617	3,519	58	1,837,988	543	4,752	60
2002	1,613,502	1,109	3,383	-136	1,843,363	5,375	4,594	-158
2003	1,625,214	11,712	3,403	20	1,846,888	3,525	4,589	-5
2004	1,614,011	-11,203	3,410	7	1,836,446	-10,442	4,562	-27
2005	1,600,108	-13,903	3,360	-50	1,824,702	-11,744	4,499	-63
2006	1,585,584	-14,524	3,353	-7	1,810,978	-13,724	4,466	-33
Total Change, 1995 to 2006	4,505		73		52,788		103	

students out of failing monopoly public schools and into charter schools? In large measure, the figures presented in Table 1 belie such a possibility. While it is true that charter school enrollments have grown substantially over time, such growth does not appear to be at the expense of enrollments in traditional public schools. Instead, the data in Table 1 suggest two fundamental sources of growth feeding charter school enrollments: (1) The total number of students enrolled in Michigan schools increased by a net 52,000 students during the study period, and (2) the table shows a precipitous decline in the number of students enrolled in non-public schools—this drop in enrollment amounts to more than 42,000

students.

Table 1 shows that net enrollment in traditional public schools edged up over the study period. During the latter '90s, enrollments increased sharply in the public schools in conjunction with simultaneous growth in charter school enrollments. Likewise, the total number of public schools in operation increased. Rather than attracting students away from the public schools, it is apparent that both school types benefitted from the strong Michigan economy of the latter '90s, which attracted younger families to the state.⁵ On the other hand, both enrollments in and number of non-public schools fell significantly during the same time period. Table 1 shows that despite the strong state economy of the latter half of the '90s and the overall significant growth in total enrollments, non-public school enrollment dropped by more than 3,300 students and 20 non-public schools closed their doors through academic year 1999–2000.

The recession of 2001, though comparatively mild, hit manufacturing states such as Michigan particularly hard. The state unemployment rate gravitated up to approximately 7.0% between 2003 and 2006. Employment growth tapered off considerably and was even negative in 2003 (at -.4%) and in 2006 (at -.2%). Population growth in Michigan in these later years slowed considerably, dropping to about +.3% from 2002–2004, followed by negligible growth in 2005 (+2,000 people), with a drop of almost 1/10 of 1% in 2006—a little bit less than 10,000 people.⁶

Non-public school enrollments begin to decrease at a significant and accelerating rate beginning in 2001 through to the end of the period. Despite the poor economic conditions and stagnating population growth, however, charter school enrollments continued to increase in the early and middle 2000s. Moreover, the gains in charter enrollments during this later period (at 43,763 students) nearly match the lost enrollment (of 38,804 students between 2001 and 2006) experienced by the non-public schools.

Is it the case, then, that charter school enrollments simply reflect a migration of students from non-public schools? Unfortunately, the data cannot answer this question directly as the data do not follow individual students longitudinally. We think, however, that the existing evidence does conform to this hypothesis. The recession of 2001 (and its aftermath) provides something of a natural

5. Economic conditions in Michigan were robust throughout the second half of the 1990s. The state unemployment rate during this period was as low as 3.7% and never exceeded 5.0%. (U.S. Department of Labor, Bureau of Labor Statistics: <http://data.bls.gov:8080/PDQ/outside.jsp?survey=la>.) Employment growth generally ranged between 1.5% and 2% per year. As a consequence of these positive economic conditions, population growth in Michigan was relatively strong at about 1/2 of 1% per year (approximately 50,000 people per year) from 1996–2000. (U.S. Department of Commerce, Bureau of Economic Analysis, Regional Economic Information System--<http://www.bea.gov/regional>.)

6. See <http://data.bls.gov:8080/PDQ/outside.jsp?survey=la> and <http://www.bea.gov/regional>.

experiment. Prior to 2001, non-public school enrollments were falling despite a strong state economy and a growing population. Did those former non-public school students transfer into the public schools or into charter schools? It is our contention that public and non-public schools are not close substitutes. Parents choose to send children to non-public schools in the first place because they think something is wrong with their local public school, e.g., no religious education on offer, a too diverse student body, and so forth. That choice is clearly not taken lightly as public schools are tuition free, whereas tuition at non-public schools is a significant out-of-pocket expense. On the other hand, charter schools are much more akin to non-public schools in terms of overall objectives and approach to education. All that they lack is the religious element,⁷ which is not necessarily important to all parents as motives for sending children to non-public schools extend well beyond religion. Charter schools, viewed in this light, are a much closer substitute to non-public schools than are traditional public schools. Because they are tuition free, charter schools have a clear price advantage over non-public schools.⁸

With the establishment of charter schools in Michigan in the middle 1990s and prior to 2001, we observe growing enrollments in *both* charter schools and traditional public schools. Such growth took place, as one would expect, with an economic backdrop of a strong state economy and a rising population. Yet, during the same period, enrollments decrease in the non-public schools. Our inference is that parents at the choice margin, when confronted with the new and free educational option, chose to move their children out of non-public schools into charters.

Post 2001, the Michigan economy slumps and population growth evaporates and is even negative in some years. Now, families sending children to non-public schools are not only at a price disadvantage vis-à-vis charter schools and traditional public schools, but also presumably experience an adverse income effect induced by the weak economic conditions in the state. We contend that rather than switch children into a traditional public school, some families that had selected into non-public schools in the first place now have a stronger

7. An anonymous referee points out that another possible explanation of the declining number of non-public schools and non-public school students is that formerly non-public schools have re-invented themselves as charter schools. There is evidence that this has occurred in some states, leading to the criticism that charter schools are a way of drawing public funding for religious schools. The enabling legislation in Michigan, however, prohibits charters from affiliation with religious organizations. Specifically: "...a public school academy shall not be organized by a church or other religious organization and shall not have any organizational or contractual affiliation with or constitute a church or other religious organization." (See Chapter 380, Section 502 (1), Michigan Compiled Laws.) Unfortunately, our data do not allow us to determine the degree to which this proscription has been complied with in practice.

8. Stoddard and Corcoran (2007: p. 44) note "...private school parents advocating public charter schools as a substitute for private schools..."

incentive to switch children into a charter school, which, at least in perception, is philosophically closer to a private school. The data in Table 1 show that the pace of exit from non-public schools both accelerates in the post-2001 period and approximately matches the gains in enrollments experienced by the charter schools after the recession. The matching of enrollment losses by the non-public schools to the gains experienced by the charters in the post-recession period is significant in that in the post-period population growth can no longer be part of the explanation of growing charter school enrollments as it would have been prior to 2001. In other words, after 2001, charter school enrollments can only grow at the expense of public and non-public schools.

According to Table 1, enrollments in traditional public schools continue to increase in the first couple of years following 2001 despite the economic downturn. Public school enrollments only begin to decline in 2004. At this point, it brings to question if some families decide whether or not to switch children out of public and into charter schools. The underlying economics certainly do not support this explanation. Charter schools possess neither a price advantage over traditional public schools, nor could there be any adverse income effect inducing switch from public schools to charter schools owing to the 2001 recession. Lacking substitution effect and/or income effect explanations, if it is the case that families began switching children into charter schools from public schools all of a sudden in 2004 and beyond, then one would have to explain such a shift on the basis of an unexpected and sustained shock to taste. Such an explanation strikes us as neither supportable nor very likely.

A more likely explanation of the declining enrollments in Michigan public schools from 2004 and on is that these decreases simply reflect exit from the state by some families rather than a migration out of public schools and into charter schools. It is at precisely this point that population growth in Michigan stagnates and then turns negative. As previously mentioned, the 2001 recession had a long half life in Michigan and, as a result, it is likely that some younger and more mobile families ultimately gave up trying to wait out the adverse effects of the recession and chose to leave the state.⁹ We contend therefore that the declines in public-school enrollments largely reflect an exit from the state rather than a transfer of students from public schools to charter schools.

This is not to say that no such transfers from public schools to charter schools took place, nor that enrollment declines among non-public schools reflect exclusive transfer from non-public schools to charters. Certainly from 2004 and on, some families with children enrolled in non-public schools left the state

9. Exit didn't occur immediately with the recession of 2001. Unemployment insurance benefits reduce mobility until they are exhausted. Furthermore, finding a new job in another state and re-locating a household impose significant search costs, which also delay mobility.

as a result of the aftermath of the recession and, no doubt, some families with children enrolled in charter schools also left the state. But the general pattern of enrollment changes, captured in Table 1, are consistent with rising enrollments in public schools when population is growing and falling enrollments in public schools when population change is below its trend and when it is negative. Non-public school enrollments begin to decline with the introduction of charter schools in Michigan and then these losses accelerate with the natural experiment of the 2001 recession. The pattern of change in non-public school enrollment is entirely consistent with a substitution effect/income effect accounting of events. Finally, charter school enrollments grow both pre and post 2001. The pre-2001 growth is explicable both in terms of population growth and in terms of flow of students out of non-public schools into charter schools. The growth in charter enrollments after 2001 is not explainable on the basis of growing population, but does match to good degree the decline in non-public school enrollments that took place in the same period.

Composition of Enrollment

In the section on *Regression Analysis*, we attempt to identify *ceteris paribus* differences between charter schools and public schools in funding per student and expenditure per student. It is important, therefore, to control for revenue and expenditure drivers to the extent that they systematically differ across school type. In this subsection, we compare school types on the basis of grade level enrollments (K–5, 6–8, 9–12) and demographic composition, as it has been argued that both grade composition and demographic composition can account for some differences in revenue and expenditure.

Table 2 presents the distribution of students by school level—elementary (grades K–5), middle school (grades 6–8), and high school (grades 9–12)—for charter schools, public schools, and non-public schools. Table 2 shows that the percentage of charter school students that attend elementary school is much greater than the percentage of public school students attending elementary school. The difference—at 10 to 15 percentage points, depending on year—is striking. We regard the enrollment figures for public schools as reflective of the age distribution of school age children in the population. That the percent of students in grades K–5 is so much higher in charter schools must reflect a conscious choice by charter school operators to serve this particular segment of the market. Comparison of charter schools to public schools at the higher grade groupings reveals an offsetting difference in percent enrolled at grades 9–12. Public school districts serve proportionally at least twice as many high-school level students as do charter schools during the study period. Michigan

Table 2. Enrollment Composition by School Type (in percent).

Year	Grades K-5				Grades 6-8				Grades 9-12				Eligible for Free Lunch		
	Charter	Public	Non-Public		Charter	Public	Non-Public		Charter	Public	Non-Public		Charter	Public	Host Districts
1996	48.7	48.7	55.1		22.0	23.2	23.2		29.3	28.1	21.7		11.6	25.1	46.8
1997	57.0	48.4	54.5		22.1	23.2	23.2		20.9	28.4	22.2		17.3	25.8	45.9
1998	62.6	48.0	54.4		21.1	23.3	23.3		16.3	28.7	22.3		20.2	25.7	44.3
1999	64.3	47.6	54.3		21.4	23.4	23.4		14.3	29.0	22.3		20.8	25.1	41.6
2000	66.8	47.2	54.3		20.5	23.3	23.2		12.7	29.5	22.6		23.6	24.9	39.1
2001	66.9	46.7	53.9		21.8	23.5	23.3		11.4	29.8	22.8		28.6	23.9	34.6
2002	63.5	45.6	52.9		23.8	24.1	23.6		12.7	30.3	23.5		29.3	25.2	36.5
2003	62.1	44.7	52.1		25.1	24.5	23.9		12.9	30.8	24.1		32.6	25.3	34.8
2004	61.1	44.3	51.9		24.8	24.6	24.0		14.0	31.1	24.1		32.9	26.6	36.5
2005	60.5	43.9	51.1		24.0	24.1	23.9		15.5	31.9	24.9		34.1	27.4	36.6
2006	59.5	43.6	51.2		24.0	23.5	24.2		16.5	32.9	24.6		39.8	28.7	37.8

Year	African-American				Female				Enrolled in Special Education			
	Charter	Public	Host Districts		Charter	Public	Host Districts		Charter	Public	Host Districts	
1996	61.5	16.9	51.3		45.6	47.7	46.5		14.8	11.3	12.1	
1997	45.4	17.3	47.0		50.1	48.0	47.3		7.6	11.4	11.8	
1998	50.8	17.4	44.4		51.1	47.9	47.5		7.5	11.8	12.2	
1999	53.8	17.5	43.0		49.7	47.8	47.4		8.1	12.2	12.7	
2000	55.5	17.7	40.1		50.3	48.9	49.4		7.5	12.6	13.0	
2001	56.6	17.2	36.0		50.7	47.8	47.6		7.8	13.0	13.6	
2002	55.2	17.2	35.1		49.4	47.6	47.2		8.3	13.3	13.8	
2003	50.5	17.6	35.5		46.7	46.9	46.4		8.5	13.1	13.5	
2004	54.2	18.0	35.4		49.9	47.5	47.5		8.6	13.5	14.1	
2005	55.2	17.5	32.8		50.6	47.6	47.6		8.8	13.9	14.5	
2006	55.2	17.7	32.6		50.3	48.0	48.0		9.4	14.1	14.7	

Source: Tabulated from Center for Educational Performance and Information data, Michigan Department of Education, available at <http://www.michigan.gov/cepi>. Note that charter school reporting of demographic characteristics is initially incomplete during the study period. In academic year 1995-1996, only 75% of charter schools report information on gender, race, and free lunch status. This figure increases to 85% the following year. From 1997-1998 and beyond, the reporting rate varies from 90-100%. Charter school reporting on special education status of students is subject to more significant non-reporting. In 1995-1996, only 6.8% of charter schools report on special education status. The charter school report rate for this variable only exceeds 50 percent beginning in 2002. The charter school report rate on special education enrollments is 86% in the last year of the sample.

charter schools clearly do not sort randomly by grade level and the data reveal a clear preference by charters to serve much younger students. To the extent that the costs associated with educating elementary-level students differ from costs incurred in educating older students, it will be important to control for grade-level composition in the regression work below.

Table 2 does not reveal much difference between the three school types in percent enrolled in grades 6–8. However, Table 2 does reveal that non-public schools also enroll proportionally fewer elementary school level children and proportionally more high-school level students than do charter schools—though the differences between these two school types are not as pronounced as the differences between charter schools and public schools. One could not say that non-public schools are neutral to the overall distribution of school-age children either. It does appear that non-public schools self select toward serving lower grades also—though the degree of this selection is not as pronounced as it is among the charter schools.

The remainder of Table 2 compares charter schools, public schools, and host-district public schools on the basis of school lunch eligibility, racial composition, gender composition, and special-education enrollment.¹⁰ Considering racial composition first, the percentage of students enrolled in charters that are African American generally falls in the 50–55% range. This figure is well above that for all public schools, which never exceeds 18% during the study period, and is also well above the percentage of students in host district public schools that are African American—which is on the order of 35% in the latter part of the study period.¹¹

How to account for the large racial gap in charter school enrollments in Michigan? The charter school movement is, of course, a school-reform effort. Such language masks a deeper reality, which is that only some public schools require “reform.” Many public school districts achieve outstanding educational results. Charter schools attempting to gain a foothold in localities with flourishing public schools are no doubt at a competitive disadvantage and are unlikely to attract students successfully in such circumstances. On the other hand, charter schools do presumably have something to offer in districts with poorly functioning public schools. If poorly functioning public schools are disproportionately located in large central cities and in urban areas in general with larger African American populations, then it is unsurprising that enrollments in

10. The Michigan Department of Education does not report demographic data for non-public schools. Host districts are public school districts containing at least one charter school.

11. In Texas in school year 2001–2002, the percent of African American students in charter schools was almost 40% and in public schools was only 14% (Grosskopf, Hayes, and Taylor 2009, p. 69). See Stoddard and Corcoran (2007) for additional discussion of the racial composition of charter schools.

charters reflect local population rather than the statewide population.¹²

Since the incidence of poverty is higher among minorities, it might be expected that higher proportions of charter school students would be eligible for the free lunch program under the National School Lunch Act. This expectation is at odds, however, with the numbers reported in Table 2. Early in the period, the percent of charter school students eligible for free lunch is below the rate in all public schools and is well below the rate in host district public schools. Miron and Nelson (2002, p. 77) point out that participation in the free lunch program is only *mandatory* for districts/charter schools offering grades K–12. Few charters offer all grades and, therefore, participation in the free lunch program is largely voluntary for charter schools. So even though charter students may have been disproportionately deserving, charter school administrators may have avoided program participation early on in an effort to avoid incursion of any costs associated with this service. For whatever reason, the percentage of charter school students eligible for free lunch comes into line with the percentage of students eligible in host-district public schools during the 2000s. This conversion may reflect parental pressure over time on charter school administrators to provide services that have come to be expected.

The percentage of female students in charter schools is about 50%.¹³ Comparison of charter schools to all public schools and to host district public schools reveals that the percent of females enrolled in charters is typically two percentage points greater than that in host-district public schools and in all public schools. While seemingly not a large difference, the persistent gap does

12. Of the 1,650 charter school-year observations in our data set, 368 of these observations (or 22.3%) are contributed by charter schools located in the city of Detroit. Similarly, Detroit charter schools from 1999 and on annually account for at least 29.6% of total charter school enrollments in Michigan. Recognizing that city borders are porous and that charter schools that vie for Detroit students do not necessarily operate in the city itself, we also tabulate school and enrollment counts for Wayne County (in which Detroit is located) and for the tri-county area of Wayne, Oakland, and Macomb counties (which comprise the inner ring of the Detroit metropolitan area). Within Wayne County from 1999 and on, anywhere from 33.9% to 35.9% of all Michigan charters operate within the county. During this same time period, the percent of all Michigan charter school students enrolled at Wayne County charters ranges between 41.8% and 45.2%. Looking at the larger tri-county area of Wayne-Oakland-Macomb, we find that the number of charters steadily increases from 41.3% in 1999 to 49.8% in 2006. Charter enrollments in the tri-county area account for 62% of the entire Michigan charter school population by the end of the period. Clearly, charter schools and charter school enrollments are concentrated in the Detroit area, hence, the comparatively large ratio of African American students attending charters cited in Table 2 reflects to significant degree the larger African American population in the Detroit metropolitan area. (In 2006, the percent of traditional public school students that is African American is 90.2% in Detroit, 46.6% in Wayne County, and 29.5% in the tri-county areas.) Source: author tabulations from the main data set.

13. Note, however, that the percent female is only 45.6% in the first year of the study period. As the note to Table 2 points out, we can't know whether the 45.6% figure accurately reflects female enrollments in charters because only 3/4 of the charters report demographic information in this particular year. It may be the case that non-reporters also had heavier female enrollments and, therefore, the 45.6% figure understates female enrollments in charters in 1995–1996.

strike us as noteworthy. The evolution of charters in Michigan has led to a more female student population in charters vis-à-vis public schools. Perhaps this disparity exists because parents are more inclined to enroll girls in charters than in traditional public schools as a way of shielding female children from the harsh environment of urban public schools. Alternatively, the disparity may have arisen from charters seeking out and catering to female students—perhaps because girls are easier or less costly to educate and, therefore, are more “profitable.”

Table 2 also breaks enrollments in special education down by type of school.¹⁴ The table shows that charter schools significantly under-serve special education students when compared against all schools and when compared with host district public schools. The percentage of special education students is slightly higher on average in host districts compared to all schools—with an incidence of approximately 14% by the end of the period. On the other hand, Table 2 shows that the percent of special education students enrolled in charter schools is relatively low throughout the period, varying between 7.5% and 9% most years.¹⁵ Special education students are costly to provide services to and Michigan charter schools may attempt to systematically avoid enrollment of these expensive students. Alternatively, perhaps parents of special education students are reluctant to enroll children in charter schools if they perceive the quantity and quality of services to be superior at the traditional public school.

In this section, a number of intriguing facts have been identified concerning Michigan charter schools. First, enrollments in Michigan charters have grown at a robust rate—both before and after the 2001 recession. Prior to the 2001 recession, most of the growth is explicable in terms of population growth, though there is also an element of charter enrollment growth that occurs at the expense of enrollment in non-public schools. Following the 2001 recession, population growth is no longer relevant as an explanatory factor and charter school enrollments in this later period reflect a large transfer of students out of non-public schools. Second, charter schools are very different from public schools in terms of grades taught. Michigan charter schools are disproportionately focused on educating students at the elementary level. Third, charter school students are demographically different from students in traditional public schools. Charter

14. These data were obtained at: http://www.michigan.gov/cepi/0,1607,7-113-21423_30451_37305--,00.html.

15. The note to Table 2 indicates that the problem of non-reporting of demographic characteristics by charters is particularly pronounced for the special education enrollment data. For example, only three of 44 charter schools report special education enrollments for 1995–1996. The reporting rate improves to 46.7% by 1999–2000 and improves further still to 87.7% by the end of the period. Given the high non-report rates for charter schools in this dimension, we do not have great confidence that the percentages stated for charters in Table 2 are unbiased. If it's the case that those charters that do report special education enrollment are also the most likely to serve special education students in the first place, then the percent enrollment figures quoted for charters with respect to special education are biased up.

school students are more likely to be African American (even compared to students in host districts), female, and about as likely to be impoverished (as proxied by free lunch eligibility) as students in host district public schools. However, they are much less likely to be enrolled in special education than students in traditional public schools.

SCHOOL REVENUES

One of the four major questions we wish to address in this article is whether, as charter school proponents often opine, charter schools are significantly underfunded vis-à-vis traditional public schools. How this question gets answered is often driven by the political agenda of the analyst. Our intent, however, is to simply document funding differences between the school types—both in total and by source—local, state, and federal. This is a necessary first step in answering the underfunding question. It may well be the case that charters are underfunded compared to public schools on average, but it may also be true that the lower funding is justified if public schools serve a more costly-to-educate population of students. As previously mentioned, charter school students are more female, African American, younger, and less likely to receive special education services. It will be important, therefore, to control for the effects of relevant education cost drivers once average differences in funding levels are determined in order to meaningfully answer the underfunding question. In other words, charter schools may be underfunded simply because they cherry-pick students vis-à-vis the local public school district, which is obligated to accept all comers.

Revenue Sources

Since the passage of Proposal A in Michigan in 1994, the major revenue source for public schools and, to an even greater degree, charter schools is the state government.¹⁶ The other two major sources of school funding are local

16. Proposal A was a statewide ballot proposition advanced by Governor John Engler. Proposal A's objective was to simultaneously "reform" both the state-tax structure and the state's system of education finance. Prior to passage of Proposal A, school funding was predominantly driven by local property taxes. For example, in academic years 1990–1991 to 1993–1994, school funding from local sources in Michigan averaged 66.3% of total revenues, while school funding from the state averaged 29.3% of the total. (Source: authors' tabulations using school district level data from the Michigan Department of Education.) Local finance of schools resulted in heavy property tax burdens and inequality across districts in revenues available per student. The inequity in funding per student was driven both by Tiebout sorting—i.e., parents that were more education focused tended to move to districts that spent more per student—and, of course, by pure income inequality. Districts inhabited by less wealthy families tended to spend less per student. Proposal A greatly reduced, though it did not eliminate, school funding from local property taxes. In place of property taxes, Proposal A substituted funds derived from state sources. Table 3a shows that the relative proportions of funds financed by state and local

government and the federal government. In Tables 3a and 3b, we present a summary of mean per student total revenue as well as mean per student revenue from the three major funding sources, set both in real terms—deflated by academic year CPI, base years 1982–1984—and in nominal terms.

Though we mainly couch our discussion of revenues and expenditures in inflation adjusted terms, we note at the outset of this section that Table 3b shows both total revenue and state revenue per student generally trended up in nominal terms for both school types throughout the study period. Comparison of Tables 3b and 3a (set in real terms), however, reveals a breakpoint after the 2001–2002 school year when the revenue time series begin to trend down in real terms, implying that funding sources stopped keeping pace with inflation in the early and middle 2000s in Michigan. This general stagnation in funding per student, regardless of school type, presumably reflects the dismal economic conditions experienced by the state in the years following the recession of 2001.

Closer study of Table 3a reveals several phenomena of note. First, mean total revenue per student in charter schools initially decreases up until 1998–1999. In subsequent years, mean total revenue per student in charter schools rises until 2003–2004, after which total revenue per student again drops off in real terms. By the end of the study period, one could not argue that charter schools were better funded in real terms than they were at the outset of the study period—\$4,282 per student in 1996–1997 versus \$4,300 per student in 2005–2006.

Examining individual revenue categories, the upward and downward movements in total per student revenue mainly reflect changes that occurred in state per student revenue.¹⁷ The contribution of the federal government to charter-school financing is relatively small. We observe a modest increase in its share from 6.8% in 1995–1996 to 7.6% by period's end. The share of local government as a source of charter school funding is even smaller than that of the federal government. There is no obvious trend in the local share, as it hovers around 3% of total per student revenue for charters throughout the study period.

Turning to traditional public schools, real per student total revenue trends up from the outset of the study period until 2001–2002, where it peaks at \$4,688. In subsequent years, the trend is generally down, falling to \$4,463 by 2005–2006. The trends in mean total revenue per student for public schools are also mostly driven by the state's ability to support public education. At the source level, the mean revenue per student from the state going to public schools rose between 1995–1996 and 2001–2002, but falls thereafter. Even though revenues from the

sources basically flip-flopped from what they were in the years leading up to passage of the proposal in March of 1994.

17. Ups and downs in the State of Michigan funds available for K-12 schools are more reflective of state budgetary problems than any explicit policy changes during the period.

Table 3a. Mean Per Student Revenue (inflation adjusted): Total, State, Federal, and Local Government.

Year	Charter Schools								Public Schools							
	Total		State Revenue		Federal Revenue		Local Revenue		Total		State Revenue		Federal Revenue		Local Revenue	
	\$	\$	%	\$	%	\$	%	\$	\$	\$	%	\$	%	%	%	%
1996	4068	3495	85.9	275	6.8	87	2.1	4193	3152	75.2	168	4.0	735	17.5		
1997	4282	3590	83.9	130	3.0	154	3.6	4228	3163	74.8	195	4.6	749	17.7		
1998	4044	3700	91.5	180	4.5	161	4.0	4381	3310	75.6	201	4.6	738	16.9		
1999	3873	3626	93.6	144	3.7	100	2.6	4341	3220	74.2	219	5.1	755	17.4		
2000	4028	3746	93.0	177	4.4	100	2.5	4387	3246	74.0	222	5.1	777	17.7		
2001	4221	3819	90.5	247	5.9	108	2.6	4480	3333	74.4	208	4.6	771	17.2		
2002	4403	3945	89.6	284	6.5	99	2.3	4688	3465	73.9	233	5.0	781	16.7		
2003	4400	3934	89.4	330	7.5	72	1.6	4599	3369	73.3	240	5.2	772	16.8		
2004	4499	3835	85.2	331	7.4	116	2.6	4578	3280	71.6	298	6.5	958	20.9		
2005	4403	3751	85.2	348	7.9	145	3.3	4588	3173	69.2	302	6.6	1008	22.0		
2006	4300	3676	85.5	328	7.6	140	3.3	4463	3121	69.9	287	6.4	1018	22.8		

Note: School and school district observations are weighted by number of students in calculating means. Data are drawn from the annual Bulletin 1014 database. Revenue values are deflated by the consumer price index tabulated on an academic year basis (base year = 1982–1984). In calculating *Total Revenue* for an academic unit, the Michigan Department of Education also counts revenue from intermediate units of government, from incoming transfers, and from other types of transactions. These other revenues are distinct from state, local, and federal revenues and, as a result, sub-categories in the table do not add up to *Total Revenue*.

Table 3b. Mean Per Student Revenue (current \$): Total, State, Federal, and Local Government.

Year	Charter Schools				Public Schools			
	Total Revenue	State Revenue	Federal Revenue	Local Revenue	Total Revenue	State Revenue	Federal Revenue	Local Revenue
1996	6301	5414	426	134	6495	4883	260	1138
1997	6821	5719	207	246	6735	5038	311	1194
1998	6551	5994	292	261	7097	5362	326	1196
1999	6383	5976	238	165	7154	5306	361	1245
2000	6839	6361	300	171	7449	5512	377	1319
2001	7407	6703	434	189	7863	5849	365	1354
2002	7850	7035	507	177	8359	6179	416	1393
2003	8031	7179	602	132	8393	6148	439	1408
2004	8391	7152	618	217	8538	6117	555	1787
2005	8459	7206	668	279	8813	6095	580	1936
2006	8583	7338	655	279	8908	6229	573	2032

other two sources generally increase over the study period, these increases were not enough to completely cancel the effect of declining state contributions.

Directly comparing charter schools with public schools by year reveals total revenue per student is generally higher in public schools, supporting the claim of underfunding of charter schools. The difference in total revenue per student averages 5.3% during the study period.¹⁸ On the other hand, Table 3a shows that per student revenue from the state is greater for charter schools, with the difference averaging \$480 per student in real terms during the entire period, and averaging more than \$535 per student from 2001 and beyond. This funding gap at the state level has important implications for the state education budget.

A substantial difference exists between charter schools and public schools on the basis of relative shares of revenue from the major sources. While the state is the largest source of revenue for both types of school, the state's share of total revenue provided is much greater for charter schools. Up until 2002–2003, the share of revenue contributed by the state to public schools is almost 75%. After this year, however, the state's share in public school revenue drops off, falling below 70% by 2004–2005. On the other hand, the share of revenue per student received by charter schools from state government is much higher than public schools, never dropping below 83% and exceeding 90% in several years.

The share of per student revenue in public schools received from local government is 17.5% in 1995–1996 and this increases to almost 23% by 2005–2006. This is a substantially higher percentage of revenue compared to that received by charter schools from local government. For either type of school, the contribution of the federal government in total revenue is relatively small. The share of per student revenue from the federal government received by public schools falls between 4.0% and 6.6%, while that received by charter schools ranges between 3.0 and 7.9% over the study period.

Altogether, Michigan charter schools on average received a bit less total revenue per student than did Michigan public schools, but charters received somewhat more in per student revenue from the state compared to traditional public schools. The natural question is why do these differences prevail? Charter schools tend to enroll fewer special education students, limited English proficiency (LEP) students, Title I students, and so forth. To the degree such students are more expensive to educate, the funding gap between charters and public schools may be completely explainable. Table 2 also showed that charter schools tend to disproportionately enroll students in grades K–5. To the extent that middle-school students and high-school students are more expensive to educate, it again follows that the total funding gap may be accounted for by

18. The average dollar difference in total revenue per student during the entire period is \$218 per student.

compositional differences in the distribution of students across grades by school type.

There are no doubt many other compositional and characteristic differences between charter schools and traditional public schools that can help explain the per student funding gaps evident in Tables 3a and 3b. The interesting question is whether a funding gap between school types remains, once the compositional and characteristic differences are taken into account. If the answer is yes, then such a difference reflects a *pure* charter school effect in funding. The next section conducts multivariate regression analyses of both per student revenue and expenditure determinants.

Budgetary Implications of the State Funding Gap

This analysis identifies a significant positive gap in funding per student between charters and public schools from the state. For example, in 2005–2006 charter schools received \$3,676 per student from the State of Michigan, whereas the typical public school district received only \$3,121 from the state. Establishment of the charter school sector in the State of Michigan induced a large-scale migration of students out of the non-public schools and into the charter schools. Migration of students from non-public schools to charter schools has an important implication for the state education budget. Michigan does not pay for or subsidize non-public school students. Tuition for these students is paid out of pocket by parents. However, the state *does pay* for students attending charter schools. Thus, for every student choosing to attend a charter school rather than a non-public school, the state must make a payment to the charter school.

Using Table 3a, the mean subsidy paid by the state to charter schools per student is \$3,676 in 2005–2006. At the same time, there are no new revenues that the state can collect, nor is there any saving in expenditure to be absorbed due to the change in a student's choice from the non-public to the charter sector. Therefore, for every 1,000 new students in charter schools, the state needs an additional budget of almost 3.7 million inflation-adjusted dollars as of academic year 2005–2006. For 42,000 additional students, this adds up to an extra \$154 million—almost 3% of the total 2005–2006 state education budget.

The higher cost per student to the state of educating a charter school student relative to a public school student should be taken into consideration. In 2005–2006, mean state revenue per student for public schools is \$3,121—implying a funding gap of +\$555 between what the state spends per charter school student versus what it spends per public school student. The total amount of increased state expenditure due to this latter funding gap is more than \$26 million dollars (in real terms). Thus, the migration of students from either public schools or

non-public schools to charter schools implies additional burden on the state budget.¹⁹ The clear implication of the analysis is that states that push in the direction of charter school reform must look for additional revenues if they don't wish to lower the revenue resources available for all public schools.

EXPENDITURES ON TOTAL INSTRUCTION AND BASIC INSTRUCTION

Determining whether a gap exists between charters and public schools will help answer the question raised in the Introduction—do charters behave more like profit-maximizing businesses than public-good providers?

Table 4 presents inflation adjusted per student expenditure on Basic Programs, Added Needs, Total Instruction, and Business and Administration for the years 1995–1996 to 2005–2006.²⁰ In the early years, 1995–1996 to 2001–2002, charter school spending on basic programs changes at annual rates ranging between -5.6% and 3.4%. Since 2002–2003, however, annual growth rates are negative—falling between -2.6 % to -5.1%. On the other hand, annual growth rates of basic program expenditures in both host district and in all public schools are generally positive. Looking across the entire 11-year period, per student expenditure on basic programs for charter schools in 2005–2006 is 11.5% lower than in 1995–1996, while per student expenditure on basic programs rose by 13.1% in host districts and by 9.5% in all public schools. Though host district basic programs expenditure is roughly 5% greater than that of charters at the beginning of the study period, this gap grew steadily over the ensuing 11 years to a difference of more than 33% by period end. Therefore, the evidence suggests that charters deliver considerably less in the way of basic programs instruction than do comparable (i.e., host district) public schools. This may be evidence that, though public in name, charters act like profit-maximizing businesses striving to minimize production cost at a given level of output per student served. The higher spending on basic programs instruction per student by the public schools is suggestive of entities acting to maximize output delivered—given a budget constraint, which is presumably the desired organizational behavior when it comes to educating children.

19. The incremental value needed by the state to pay for students in charter schools in total is \$180 million, or about 3.4% of the total state education budget.

20. Per Bulletin 1014, Basic Programs expenditures consist of classroom costs associated with instruction for preschool, elementary, middle, and high school programs. Added needs expenditures include the classroom costs associated with special education and with compensatory education. Total Instruction expenditures consist of basic instruction expenditure plus expenditures for added needs and adult education students—neither of these expenditure categories include capital outlay. Business and Administration expenditures include the cost of general administration, school administration, business services, and central services.

Table 4 also reveals a stunning gap between what charter schools and public schools spend per student for special and compensatory education. As nominal public schools, charters are subject to the same rules and regulations as traditional public schools regarding provision of special education services. However, the added-needs expenditure figures reinforce the picture of considerable differences between traditional public schools and charter schools suggested by

Table 4. Real Per Student Instructional Expenditures: Basic, Added Needs, and Total.

Year	Basic Programs Instructional Expenditure						Added Needs Instructional Expenditures					
	Charter Schools		Host District Public Schools		All Public Schools		Charter Schools		Host District Public Schools		All Public Schools	
	Mean	%Δ	Mean	%Δ	Mean	%Δ	Mean	%Δ	Mean	%Δ	Mean	%Δ
1996	1709		1783		1861		131		824		522	
1997	1737	1.6	1809	1.5	1889	1.5	119	-9.3	805	-2.3	538	3.0
1998	1719	-1.0	1795	-0.8	1868	-1.1	96	-19.2	781	-3.0	544	1.1
1999	1623	-5.6	1838	2.4	1902	1.8	80	-16.4	795	1.8	564	3.7
2000	1657	2.1	2008	9.2	1975	3.9	124	54.1	756	-5.0	579	2.6
2001	1701	2.7	2009	0.1	1987	0.6	188	52.0	787	4.1	611	5.6
2002	1760	3.4	2010	0.1	2021	1.7	231	23.0	742	-5.7	599	-2.0
2003	1670	-5.1	2004	-0.3	2026	0.3	290	25.1	741	-0.1	603	0.6
2004	1609	-3.7	2093	4.4	2072	2.2	286	-1.3	764	3.0	609	1.0
2005	1553	-3.5	2097	0.2	2077	0.3	296	3.6	721	-5.6	595	-2.2
2006	1512	-2.6	2017	-3.8	2037	-1.9	286	-3.5	702	-2.7	588	-1.3
Average Annual %Δ		-1.21		1.24		0.91		8.09		-1.60		1.19

Year	Total Instructional Expenditures						Business and Administration Expenditures					
	Charter Schools		Host District Public Schools		All Public Schools		Charter Schools		Host District Public Schools		All Public Schools	
	Mean	%Δ	Mean	%Δ	Mean	Growth Rate	Mean	Growth Rate	Mean	Growth Rate	Mean	Growth Rate
1996	1840		2725		2458		922		545		472	
1997	1856	0.9	2671	-2.0	2473	0.6	1018	10.4	533	-2.2	477	1.2
1998	1816	-2.2	2630	-1.5	2457	-0.6	1132	11.2	528	-0.9	482	1.0
1999	1703	-6.2	2687	2.1	2510	2.1	1012	-10.6	585	10.7	509	5.7
2000	1781	4.6	2809	4.6	2595	3.4	1364	34.8	547	-6.4	509	0.0
2001	1890	6.1	2841	1.1	2639	1.7	1225	-10.2	545	-0.4	513	0.9
2002	1991	5.4	2795	-1.6	2662	0.9	1221	-0.4	639	17.1	560	9.0
2003	1960	-1.6	2781	-0.5	2668	0.2	1149	-5.8	613	-4.0	551	-1.6
2004	1894	-3.3	2842	2.2	2676	0.3	1056	-8.2	649	5.9	570	3.4
2005	1849	-2.4	2809	-1.2	2670	-0.2	1019	-3.4	617	-5.0	556	-2.4
2006	1798	-2.8	2719	-3.2	2626	-1.6	995	-2.4	576	-6.7	533	-4.0
Average Annual %Δ		-0.23		-0.02		0.67		0.77		0.55		1.24

Note: School and school district observations are weighted by number of students. Real values were then obtained by deflating nominal values using the consumer price index tabulated on an academic year basis (base year = 1982–1984).

the special education enrollment data presented in Table 2. Public schools in general typically spend \$550 to \$600 per student on added needs instruction throughout the study period. Michigan charters, on the other hand, spent just under \$300 per student for added needs by the end of the period. The spending gap is more pronounced in the later '90s when charters spent less than \$100 per student in some years on added-needs instruction. Table 4 shows, however, that the added-needs spending gap closed to some degree since academic year 1998–1999. Nevertheless, even at the end of the study period, charters spent only 41% of what host-district schools were spending on added needs and only 49% of what all public schools were spending. As with the basic-programs expenditure data, the evidence suggests that charters deliver substantially less in the way of services in this realm.

Why do such large differences in added-needs expenditures exist? Is it that charters purposefully under-deliver special education services in order to minimize costs? Are parents of special needs children more likely to enroll their children in traditional public schools because they believe the level of support and service to be superior? Unfortunately, the data do not allow a test of these competing hypotheses.²¹ The evidence is clear, however, that Michigan charters spend much less on a per student basis in provision of special and compensatory education services. Charter school advocates may argue that charters provide cost efficiencies in this realm, but the enrollment and expenditure data suggest that the savings come at the expense of services provided.

The pattern of spending by charter schools on total instruction per student looks similar to that of per student spending on basic programs. There is some improvement in spending by charters on total instruction per student from 1998–1999 to 2001–2002, but the trend in this category is consistently down from 2002–2003 onward. By period end, charter school real expenditures on total instruction per student are actually 2.3% *less* than their level in 1995–1996. In contrast, the trend in host district and in all public school spending on total instruction is generally positive in the first half of the period and flat after 2001–2002. We note that all public schools over the years 1995–1996 to 2005–2006 increased their real total instruction spending per student by 6.8%, while there was essentially no change in per student total instruction spending by the host districts.

Comparison of charter schools and public schools clearly suggests that public schools spend more on instruction per student than do charter schools. Moreover, the gap in public school real expenditures per student on instruction

21. In fact, these mechanisms may be reinforcing rather than competing in the sense that parents self selecting into public schools may be a direct result of charters shirking their obligations in the realm of special education.

over charter schools is larger than the gap in real total revenue per student in almost every year. We calculated the proportion of per student total revenue that is spent on basic and on total instruction. Comparison of charter schools with public schools on this basis reveals that public schools allocate a larger proportion of available resources (total revenue) to basic and total instruction. This gap is larger for total instruction (ranging between 11.7% and 16.7%) than the basic instruction spending gap (which ranges from 0.5% to 9.0%).

Given that Michigan charter schools spend proportionately less out of total revenue on instruction than do public schools, the natural question is where do they spend their money? The previously mentioned proposition suggests it may be the case that, though public by definition, charter schools behave more like profit-maximizing firms than social institutions attempting to optimize on the level of service provided, given budgetary constraint. In this event, the excess revenue generated per student by charter schools might manifest as higher administrative expense—for example, in the form of higher salaries paid to administrators and school operators. Table 4 shows a clear difference between per student business and administration expenditures incurred by charter schools relative to public schools. Comparing charters to host districts, the table reveals that business and administration expenditures incurred by charters are double those incurred by host district public schools in many years of our sample period. In the last year of the period, per student business and administration expenditures by charter schools exceeded those of host district public schools by more than 72%. This expense category does not exhibit any pronounced trend over time for any of the three school types, but the difference in level between charter schools and public schools is remarkable and supports the proposition that charter schools behave more like profit-maximizing firms where the “profits” show up as administrative expense.

Summing up the results of this section, we find interesting differences in expenditures between charter schools and public schools. Specifically, public schools spend considerably more per student on both basic programs and on added-needs instruction than charter schools. Conversely, Michigan charter schools incur much greater administrative cost per student than public schools. All of these results are consistent with the proposition that charter schools behave as profit maximizers, or, less kindly, as rent seekers under the incentive structure of Michigan’s system of public education. We hasten to point out that our conclusions, thus far, are based on comparison of unconditional averages across school types and that it may be the case that the observed differences in revenues and expenditures are attributable to differences in populations served and other characteristics beyond the control of school operators, rather than the specific type of school—charter versus public. This latter issue is addressed using

regression analysis, which is discussed in the next section.

REGRESSION ANALYSIS

The descriptive analysis above shows considerable unconditional differences in revenue and expenditure per student between charter schools and public schools. For example, Table 3 shows that charter schools received \$163 less total revenue per student than did traditional public schools in 2005–2006. Likewise, Table 4 shows that public schools spent \$525 more per student on basic programs instruction than did charter schools in that same year.

Why do charter schools and public schools differ in revenue per student and in expenditure on instruction per student? The regression analysis allows us to control for various factors that determine school revenues and expenditures. According to Table 4, charters spend less per student than do public schools. It may be the case that charter schools provide less value for money than do public schools in terms of expenditure on basic instruction, perhaps because charters are motivated to retain more revenue for administrator salaries—a hidden profit. If so, then we would expect to find a pure, negative, charter school effect on spending. On the other hand, observed differences in expenditure per student (or in revenue per student) may reflect compositional differences across school types in populations served, location, scale economies, management structure, and so forth. It may turn out that charter schools are rich in characteristics and attributes generating lower per student expenditures and per student total revenues. The only way to differentiate between the two possibilities is regression analysis.

In the work reported below, we estimate regressions of the following general form: $Y_i = X_i\beta + \gamma C_i + \varepsilon_i$

where Y_i is the variable of interest (e.g., revenue per student, basic instruction per student, etc.,) in school (or district) i ;

X_i is a vector of variables believed to determine Y_i ;

β is a vector of regression coefficients;

C_i is an indicator variable equal to 1 if the educational unit is a charter school and equal to 0 otherwise;

γ is a regression coefficient measuring the pure²² charter school effect; and

ε_i is the error term for observation i , reflecting variation in Y_i due to omitted factors.

22. We hasten to caution that in the empirical work below, the estimated γ represent our attempts to identify the “pure,” or *ceteris paribus*, effect of charter schools on revenue and on expenditure per student, holding other important factors driving per student revenues and expenditures equal across schools. The degree to which our attempts in this endeavor are successful, of course, comes down to the quality and comprehensiveness of our set of controls, X .

From Tables 3 and 4 we have rough (or unconditional) estimates of γ for revenue and expenditure per student respectively. For example, Table 3 shows charters received \$219 less per student on average over the entire study period in total revenue, but charters received an average of \$480 more in per student revenue from the state during this period. Using the data in Table 4, we calculate that charters spent an average \$315 less per student on basic programs instruction than did Michigan public schools from 1995–1996 to 2005–2006. Furthermore, the Table 4 data show that total instructional expenditures were an annual average \$732 lower for charters in comparison to public schools. The relevant question is whether these gaps reflect a pure charter effect or instead are the result of charter schools and public schools differing in the various attributes and characteristics that determine per student revenue and expenditures in general.

Revenue per Student

The unit of observation in the regression analysis is the school district for public schools and the individual school for charter schools. Our sample contains all school districts as well as all charter schools from 1995–1996 to 2005–2006. Regression results are presented in Table 5. The dependent variable in columns 1 and 2 of Table 5 is total revenue per student and the dependent variable in columns 3 and 4 is state revenue per student.²³ Explanatory power (measured by R^2) is much higher in regressions using state revenue per student as the dependent variable. Addition of county (location) fixed effects to the regression adds about .14 to the value of R^2 in the case of total revenue per student and .105 to R^2 in state revenue per student regressions.

The regressions reported in Table 5 estimate effects of a variety of factors we posit as important in determination of total and state revenue per student. We discuss results for the control variables prior to discussing the estimated charter effects. First, the effect of school size—as measured by number of students enrolled—on revenue per student is always positive in the revenue regressions and is significant in all four of the regressions. Intuitively, larger schools receive more revenue per student, though the effect is not quantitatively large.²⁴ Second, we find that the student-to-teacher ratio has a negative sign and is significant in all regressions. Thus, schools with crowded classes generally receive less revenue per student.²⁵ Third, we use two variables to control for grade level of students

23. All revenue and expenditure variables are inflation adjusted.

24. Based on the coefficient value, addition of 100 students increases per student revenue anywhere from \$.30 to \$1.20.

25. One may question the direction of causality here. That is, do large class sizes generate less revenue per student or does low revenue per student generate large class sizes? Our intent in including this variable is only to control for pupil-to-teacher ratio across schools. In other words, in estimating the

Table 5. Regression Analyses: Revenues and Expenditures per Student, 1995–1996 to 2005–2006.

Variable	Revenue per Student				Expenditures per Student			
	Mean Total Revenue per Student		Mean State Revenue per Student		Basic Programs Expenditures		Total Instructional Expenditures	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Charter School Dummy	-185.985*	-230.006*	346.917*	337.645*	-621.195*	-562.724*	-958.749*	-910.352*
	(-5.13)	(-6.48)	(18.31)	(18.40)	(-27.51)	(-24.63)	(-41.07)	(-38.42)
Number of Students	0.012*	0.006*	0.007*	0.003*	0.003*	0.002**	0.007*	0.005*
	(7.29)	(4.10)	(8.18)	(3.43)	(3.31)	(2.04)	(7.69)	(6.15)
Student to Teacher Ratio	-95.759*	-83.040*	-5.037*	-12.968*	-61.189*	-55.945*	-59.149*	-53.812*
	(-32.43)	(-27.78)	(-3.16)	(-8.44)	(-37.52)	(-32.98)	(-35.11)	(-30.64)
% Middle School (grades 6-8)	-1.172	-1.332	1.566**	2.338*	-1.605**	-1.244***	-1.279**	-1.156***
	(-0.90)	(-1.10)	(2.41)	(3.94)	(-2.30)	(-1.85)	(-1.76)	(-1.64)
% High School (grades 9-12)	3.088*	2.739*	2.616*	3.320*	2.218*	2.595*	3.203*	3.320*
	(4.50)	(4.23)	(7.69)	(10.50)	(6.04)	(7.22)	(8.45)	(8.91)
% African American	4.874*	1.030***	3.891*	0.591**	0.257	-0.760**	0.656**	0.026
	(9.19)	(1.82)	(14.77)	(2.14)	(0.90)	(-2.42)	(2.20)	(0.08)
% Hispanic	-1.875	1.528	1.421	-1.285	0.365	1.161	1.654***	3.012*
	(-1.06)	(0.83)	(1.62)	(-1.44)	(0.39)	(1.15)	(1.69)	(2.85)
% Female	-18.012*	-23.286*	-4.944*	-5.438*	-11.686*	-15.382*	-14.434*	-17.782*
	(-7.59)	(-10.5)	(-4.18)	(-4.99)	(-9.2)	(-12.49)	(-10.95)	(-13.86)
% Receiving Free Lunch	6.764*	9.102*	-1.110*	1.545*	-2.915*	-1.592*	1.971*	2.801*
	(10.69)	(14.09)	(-3.34)	(4.72)	(-8.61)	(-4.43)	(5.60)	(7.49)
Real Revenue per Student from Local Sources	--	--	-0.309*	-0.265*	--	--	--	--
			(-53.45)	(-43.96)				
Real Revenue per Student from Federal Government	--	--	0.142*	0.156*	--	--	--	--
			(8.05)	(9.03)				
Real per Student Spending on Administration	--	--	--	--	0.491*	0.400*	0.533*	0.425*
					(21.31)	(17.54)	(22.28)	(17.90)
Constant	6599.990*	6822.035*	3340.057*	3698.855*	3505.818*	3627.658*	3872.175*	4027.897*
	(45.54)	(45.78)	(44.48)	(49.06)	(43.53)	(42.55)	(46.53)	(45.50)
Year Effects	YES	YES	YES	YES	YES	YES	YES	YES
County Effects	NO	YES	NO	YES	NO	YES	NO	YES
N	7599	7599	7599	7599	7597	7597	7606	7606
R ²	0.187	0.332	0.486	0.591	0.295	0.376	0.361	0.431

Note: Revenue and expenditure values are deflated by the academic year consumer price index. Demographic variables are drawn from Michigan Department of Education, Center for Educational Performance and Information, Public Student Counts data at http://www.michigan.gov/cepi/0,1607,7-113-21423_30451---,00.html. Free lunch data are drawn from the Common Core of Data for Public Elementary and Secondary Schools, U.S. Department of Education, National Center for Education Statistics. All other variables are drawn from the Bulletin 1014 database. t-Statistics are reported in parentheses.

* ~ significant at the 1% level
 ** ~ significant at the 5 % level
 *** ~ significant at the 10% level

taught—% *Middle School* and % *High School*. % *Middle School* is the percent of the school's (or district's) total enrollment enrolled in grades 6–8. Likewise, % *High School* is the percent of students enrolled in grades 9–12.²⁶ As can be seen from the results in columns 3 and 4, the state allocates more resources (revenue) the higher the school level is and these observed effects are statistically significant. From the point of view of total revenue per student (columns 1 and 2), only the coefficient on % *High School* is positive and significant. Fourth, the % *African-American* variable has a positive and significant coefficient in all of the revenue regressions. Thus, government financing of schools (districts) provides more resources, other things equal, to schools with larger African-American student populations. On the other hand, the percentage of Hispanic students enrolled usually has an insignificant coefficient and an unstable sign. Fifth, the % *Female* variable has a negative and significant coefficient in all revenue regressions. Thus, schools with a higher proportion of female students get less revenue per student. Note that the magnitude of the coefficients of this variable in columns 1 and 2 is about 4 times larger than in columns 3 and 4, which may indicate that the revenue from local government is mainly responsible for this effect. Sixth, the percentage of students who receive a free lunch—a measure of poverty—has a significant and positive coefficient as expected in all regressions but column 3. The coefficients for this variable are about 6 times larger in columns 1 and 2 than in columns 3 and 4—reflecting that funding for free lunches is driven by the federal government. Finally, we also include real revenue per student from local government and from the federal government. These variables are included only in the state revenue regressions (columns 3 and 4). The coefficients of these variables are statistically significant. The sign on the local revenue variable is negative, implying some substitution between state and local revenue sources. The sign on the federal revenue variable in contrast is positive, suggesting that state and federal revenues are somewhat complementary.

The variable of prime interest in the regression analyses is the charter school dummy. The coefficients on this variable allow us to estimate the effect on per student funding (and on per student spending) of charter schools, having controlled for other revenue (and spending) determinants. Table 5, columns 1 and 2, show a negative and significant charter school effect in the total revenue regressions. Since charters received an unconditional annual average of \$219 less than public schools in total revenue per student during the study period, the regression estimates are consistent with the assertion that charter schools

effect of being a charter on per student revenue, it is important to hold class size equal in order to properly make the *ceteris paribus* comparison.

26. Percentage of students enrolled in grades K–5 (i.e., elementary school students) is the base category and is excluded from the regression to avoid perfect collinearity.

generally receive less total revenue per student than do public schools in Michigan. Moreover, given that the charter school effect estimates in columns (1) and (2) are within a few dollars of the unconditional average difference, the regression results imply that virtually none of the observed difference in total revenue per student across school types can be attributed to differences in underlying characteristics and attributes of schools.

In the state revenue per student regressions (columns 3 and 4), we find a positive pure charter effect. Charter schools are over-funded by the state relative to comparable public schools. The coefficient of the charter dummy in columns (3) and (4) is positive, statistically significant, and larger in absolute value compared to the charter coefficient estimates in columns (1) and (2). The state regressions show that Michigan pays out almost \$350 more per student to charter schools than to public schools, other things equal.²⁷

The estimated coefficient values for the charter effect in the state revenue

27. An important demographic variable that we are forced to omit from the regression work is percent of students enrolled in special education. As noted above, reporting of special education enrollments by charters is spotty throughout much of the study period and hence this variable receives a missing value for many charter schools in the data set. Because of the missing value issue, inclusion of the variable in the regressions would cause loss of more than 900 observations, most pertaining to charter schools—40% of all charter school observations are lost. If non-reporting of special education enrollment is determined randomly across schools, then running the regression with the variable included should not bias the estimated regression coefficients. As discussed above, however, it is our contention that non-reporters have systematically lower special education enrollments than do charters that report the information. Inclusion of only those charters that do report their special education enrollments, then, is problematic since the selection into the sample is not random.

What is the cost of omitting special-education enrollments as a regressor? The expected value of the regression coefficient of interest in the presence of an omitted variable is $E(\tilde{\beta}_j|X) = \beta_j + \beta_k \tilde{\delta}_j$, where $\tilde{\beta}_j$ is the partial effect of the variable of interest in the regression that omits variable k , β_j is the true partial effect of the variable of interest, β_k is the true partial effect of the omitted variable on the dependent variable, and $\tilde{\delta}_j$ is the regression coefficient of the variable of interest in an auxiliary regression of the omitted variable on all other variables in the regressor set. (See Wooldridge 2009, pgs 114–115.) In the present context, the charter dummy is the variable of interest and percent enrolled in special education in the academic unit is the omitted variable. The dependent variable is total revenue per student in regressions 1 and 2 and is state revenue per student in regressions 3 and 4.

We think it likely that the true charter effect in the total regressions is negative (i.e., $\beta_j < 0$). In other words, other things equal, charters receive less total revenue per student than do comparable public schools. Furthermore, we believe that schools that have higher special education enrollments do receive more total revenue per student other things equal (i.e., $\beta_k > 0$). Finally, the correlation between special-education enrollments and charter schools is negative; special-education enrollments are much lower in charter schools than in public schools. Hence, we expect that $\tilde{\delta}_j < 0$. Table 5 reports the estimated partial effect for charters with special-education enrollments omitted, i.e. $\tilde{\beta}_j$, as negative in the total revenue regressions. If special education enrollment has a positive partial effect on total revenue and if special education enrollments and charter schools are inversely related, then the $\beta_k \tilde{\delta}_j$ term must be negative, which implies that the charter coefficient is biased away from 0 in the negative direction in columns (1) and (2). Said alternatively, the charter coefficients in columns (1) and (2) overstate the degree by which charters are disadvantaged with respect to total revenue funding.

In contrast, we believe the true charter effect in the state regressions (columns 3 and 4 in Table 5) is positive. In other words, state funding per student is *ceteris paribus* greater for charter schools than for traditional public schools. Given positive β_j , it stands to reason that the reported $\tilde{\beta}_j$ in columns (3) and (4) of Table 5 establish a lower bound on the charter effect vis-à-vis state funding per student.

regressions are approximately the same whether or not one controls for location. However, the estimated charter effect on state revenue per student is smaller than the unconditional annual average difference of \$480 identified above. Therefore, the regression results of columns (3) and (4) imply that almost 30% of the unconditional average difference in state revenue per student is explicable on the basis of differences in underlying school characteristics. Nevertheless, most of the higher funding flowing to charters from the state is a pure charter effect. Michigan funds charter school students at a better level than it funds traditional public school students.

The introduction of charter schools in Michigan induced a major shift in enrollments out of non-public schools, where students cost the state nothing, to charter schools, where the state not only must cover the cost of educating each student, but actually pays a premium of almost \$350 per student over the rate paid traditional public school—in correspondence with the results of columns (3) and (4) in Table 5. Furthermore, to the extent that charter schools have also induced some shifting of students away from traditional public schools to charter schools, the Table 5 results imply that such shifting is not revenue neutral. Since charter schools cost the state more per student than do traditional public schools, the implications for the state education budget are clearly adverse.

Expenditure on Basic and Total Instruction Per Student

Table 5 presents the results of OLS regressions using basic programs expenditure per student (columns 5 and 6) and total instructional expenditures per student (columns 7 and 8) as dependent variables. The performance, in terms of sign and significance, of most of the control variables in the expenditure equations is as expected. The student-to-teacher ratio regression coefficients are consistently negative and highly significant. Schools and districts with larger student-to-teacher ratios spend significantly less per student on both basic programs and in total. % *High School* is positive and significant in all of the expenditure regressions. High school students are more expensive to educate. Hispanic student population has no statistically significant effect on per student spending for basic programs, but does have a positive and significant effect on total instructional expenditures. Moreover, controlling for county effects in column (8) causes a near doubling of the predicted effect, thus a proportionally larger Hispanic student population within a county leads to considerably more per student expenditure on total instruction—other things equal.

One variable with a somewhat unexpected coefficient sign is *Number of Students*, which is positive and significant in columns (5) to (8) of Table 5. This result suggests that the greater the number of students, the larger the expenditure

per student, i.e., there are no scale economies in instruction expenditure.

Estimated coefficients for % *African American* are unstable but display an interesting pattern. The coefficient is small and not statistically significant in column (5), where county/location effects are not controlled for, but it is negative and statistically significant in column (6), where location is controlled for. Other things equal, then, schools with larger African-American populations *within* a county spend less per student on basic programs. For example, the model predicts that a 10% increase in African-American students depresses expenditure per student on basic programs by \$7.60 within a county. In contrast to the results reported for this variable in columns (5) and (6), the regression coefficient for % *African-American* is positive and significant in column (7), which pertains to total instructional expenditures. County effects are not controlled for in this regression; therefore, given two otherwise identical schools or districts in the State of Michigan, the school or district with the proportionally larger African-American student population spends more per student on total instruction. Recall that total instructional expenditures include spending on compensatory education and on adult education, thus, the regression seems to be saying that a larger African-American student population leads to disproportionately higher compensatory and adult-education expenditures. Curiously, when county effects are controlled for (column (8)), the coefficient on % *African-American* is near zero and not significant. Thus, for two otherwise identical schools within a county, proportionally more African-American students has no effect on Total Instructional Expenditures.

Two other variables with unexpected signs in the expenditure regressions are % *Female* and % *Receiving Free Lunch*. Both variables have a negative coefficient and both are significant in the basic programs expenditure regressions. Schools and districts clearly spend less per student on basic programs the greater is the proportion of students that are female. This may imply that girls are less costly to educate than boys or perhaps it reflects gender bias. Likewise, schools spending less per student on basic programs, the larger the low-income population is a puzzling and also disturbing discovery. At the very least, one would hope that a largely state-financed system of public education, such as Michigan, would take steps to ensure that per student instructional expenditure is at least neutral to the income level of the students—if not biased toward eliminating such inequities, by spending more per student the larger the proportion of low-income students. In the total instructional expenditure regressions—columns (7) and (8)—the coefficient of % *Receiving Free Lunch* switches signs and is significant. Given the nature of total instructional expenditures, which include expenditures for compensatory education and for adult education, this sign switch squares with intuition. Other things equal, areas with lower income populations no doubt

require more compensatory and adult education services, accounting for the positive sign.

Having controlled for other important determinants of expenditure, the next question to ask is: What is the pure, or *ceteris paribus*, effect of charter schools on per student expenditures? According to Table 4, charter schools spent on average \$315 less per student than did public schools over the course of the study period on an unconditional basis. This gap in spending is evidence that charters act like profit-maximizing businesses and that, as a consequence, charters deliver less educational output per dollar of revenue than do traditional public schools. In the absence of regression analysis, however, one cannot know for certain whether the spending gap exists because charters are attempting to maximize profit or because charters are fundamentally different from public schools with characteristics and attributes that result in lower spending per student.²⁸

Columns (5) and (6) in Table 5 show that, upon controlling for the effects of other explanatory variables, charter schools spend between \$563 and \$621 less per student on basic programs than do public schools.²⁹ The charter school effect on basic programs expenditure is nearly twice that suggested by comparison of unconditional means. Differences in underlying determinants of instructional expenditure, therefore, serve only to mask the difference in spending that one would find in other things equal comparison. Michigan charter schools spend considerably less per student on basic programs than do Michigan public schools.

Columns (7) and (8) of Table 5 show the pure charter effect on total instructional expenditures is also substantially greater than what simple comparison of unconditional means would suggest. Using Table 4, public schools' total instructional expenditures per student were an average \$732 greater than those of charter schools over the study period. Depending on whether or not we control for county effects, however, Table 5 reveals that the true, *ceteris paribus*, differential was at least \$910. Hence, whether we use a narrower or a broader measure of instructional expenditure, the evidence is clear that Michigan charter schools spend much less on instruction per student than do Michigan public schools. The evidence here leads us to conclude that charter schools do act much like profit-oriented businesses. This strikes us as a negative outcome, as lower expenditure on instruction per student surely maps into less educational output

28. For example, elementary school students make up a disproportionate percentage of charter students in comparison to public school students and, to the extent that younger children are less costly to educate than older children, one would expect charter school spending to be lower.

29. Grosskopf, Hayes, and Taylor (2009) find that Texas charter schools spend less per student than do traditional public schools. They note that: "Although charter schools have the same number of teachers per student as other schools, they spend a significantly lower share of their budgets on instruction. In particular, charter schools hire significantly more administrators per pupil and significantly fewer teacher aides and auxiliary workers" (p. 70).

delivered by the charter schools.

CONCLUSION

Four questions were asked at the outset of this article: (1) To what degree do charter schools compete with local public schools? (2) Are charter schools underfunded, *ceteris paribus*, in comparison to traditional public schools? (3) What is the effect on state education finances of operating an essentially second system of public education? (4) Do charter schools, judging by expenditure from revenue patterns, act more like profit-maximizing businesses than like public good providers?

While there is no doubt some competition between charter schools and public schools, much of the growth in charter school enrollments during the study period occurs simultaneous with a marked decrease in enrollments of approximately the same magnitude in non-public schools. This finding is a particularly noteworthy aspect of this article and certainly worthy of future research. The phenomenon suggests that though charter schools in theory offer an alternative to public schools, the reality is that tuition-free charter schools actually serve as an attractive alternative to tuition-bearing non-public schools. Moreover, charter schools spend substantially less per student on instruction compared to public schools—suggesting that the educational output of charters falls short of that from public schools. Assuming that parents of public school students are aware of the differences in per student expenditures on instruction, this would account for charter schools having been much more competitive with non-public schools than with public schools. Despite the would-be flaws of troubled public schools apparent to ostensible school reformers, the data suggest that parents of public school students are generally aware that charters offer less educational output per dollar spent than the local public school. If this is indeed the case, then charter schools as a reform of public education miss the intended mark.

Our second question asks whether charter schools are underfunded vis-à-vis traditional public schools in Michigan. We found that, on an unconditional basis, charter schools received \$219 less in total per student revenue than did public schools on average over the entire sample period. Control for various revenue determining factors in the regression analysis yields little impact on the estimated charter effect—the estimated *ceteris paribus* charter effect is a negative \$230 in column (2) of Table 5. Therefore, it appears that charter schools are indeed underfunded on a total revenue per student basis. This may be because the state shoulders most of the responsibility for financing charter schools, whereas a fairly significant source of funding for traditional public schools

continues to derive from local property taxes. On the other hand, our inability to include special-education enrollments in the regression work most likely implies that the conditional estimated funding gap is an upper-bound estimate. That is, if special-education students are more costly to educate and if public schools have considerably greater levels of special education enrollments, then part of the observed gap in total revenue per student is attributable to the unobserved differences in special-education enrollments.

Furthermore, while charter schools may be underfunded on a total revenue per student basis, they are clearly overfunded on a state revenue per student basis. Unconditionally, charter schools received \$480 more per student from the State of Michigan than did public schools on average during the study period. Controlling for differences across school types in underlying revenue determinants reduces the estimated differential to a nevertheless substantial \$338 per student. Other things equal, the state pays out considerably more to charter schools per student than it does to public schools.

This latter point ties into the third question asked in this article—what is the effect on state education finances of operating an essentially second system of public education? Charter schools are one type of school reform. As such, it is reasonable to explore whether charters are revenue neutral with respect to the state education budget. Our analysis shows that, to the contrary, charter schools are an expensive brand of educational reform. First, the state pays out almost \$350 more per charter school student than per public school student. To the extent that the existence of charter schools has caused a migration out of traditional public schools into charter schools, this migration has a significant adverse consequence for the state education budget due to the premium the state pays for charter school students. Second, while it is debatable how large the degree of migration from public schools to charter schools has been, this article reveals a significant migration of students from non-public schools to charter schools over time. Students in this category impose the full cost of a charter school student on the state budget. We estimate that by the end of the study period, Michigan's charter school system imposed an incremental cost of \$180 million on the state's education budget via these two effects.

The fourth question weighs whether Michigan charter schools behave more as profit-maximizing businesses than as public good providers. In order to answer this question, we examine per student expenditure on instruction data. Unconditional comparisons show that public schools spend more on instruction per student than do charter schools throughout the period and that this gap grew from about 5% at the beginning of the study period to almost 35% by the end of the period. In real dollar terms, the unconditional average difference on per student spending for basic programs is \$315 between 1995–1996 and

2005–2006. The regression analysis shows, furthermore, that when underlying determinants of per student expenditure on instruction are controlled for, the spending gap between public schools and charter schools is even more pronounced. The spending gap between public schools and observationally equivalent charter schools is estimated at at least \$562 for basic programs and \$910 for total instructional expenditures during the study period. Tying this point back to the underfunding question, we note that while our regression work shows that charter schools are “underfunded” by perhaps as much as \$230 per student on a total revenue basis, they more than make up for this deficiency by spending \$562 less per student on basic programs and \$910 less per student on total instruction. In other words, much like a profit-maximizing firm, charter schools generate a surplus of revenue over expenditure. Michigan charter schools are, technically speaking, public schools, so the surplus is not recorded as profit *per se*. Rather, the surplus manifests as grossly inflated per student business and administration expenses. During the study period, average per student business and administration expenditure among charter schools is \$1,101 while the average among public schools is \$521—a \$580 gap.

Do Michigan charter schools behave more like profit-maximizing businesses than public goods providers seeking to maximize output—educational output in this instance—subject to a budget constraint? Our reading of the evidence is that the answer to this question is yes. *Ceteris paribus*, charter schools spend far less per student on instruction than do public schools. Moreover, charter school expenditure on administration, at double the per student dollar amount incurred by public schools, very much appears to be a case of profit hiding as cost.

Charter schools in Michigan were introduced as part of a “reform movement” for failing public schools. Public schools were deemed to be failing in a variety of dimensions such as low-test scores and low-graduation rates. The reality, of course, is that the majority of public schools in Michigan are not “failing” and, therefore, charter schools do not compete with these public schools. Moreover, in areas in Michigan where the public schools are failing, the explanation for the failure has much less to do with poor teaching and much more to do with underlying socioeconomic circumstances such as poverty, single-parent households, and even homelessness.

Charter schools are on some level an ideological proposition. The charter school concept follows from the point of view that less government and more competition are always to be desired. Our analysis leads us to the opposite conclusion, however. We find that charter schools mainly offer competition (and perhaps unfair competition at that) to non-public schools that had held a niche in the marketplace prior to the introduction of charter schools in Michigan. We

find, in addition, that charter schools spend considerably less on instruction than do comparable public schools and spend considerably more on administration. The inference that charter schools deliver less educational output per dollar of revenue is clear. Finally, the state spends considerably more per charter student than it does per public-school student. The implication is that the establishment of charter schools has not been revenue neutral. The state spends significantly more on education than it would in the absence of charter schools, both because it pays out a premium on behalf of a student who would otherwise attend a public school and because many students attending charter schools would otherwise have attended a non-public school.

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